# Supporting Information

A testing-effort artefact in the most-sensitive aquatic endpoint: structure-based PFAS screening recovers predictive validity but still cannot support clearance

Jeongyun Kim, Jin Yoo, Ku Kang

## S1 Extraction from the ECOTOX Knowledgebase

All toxicity data derive from the U.S. EPA ECOTOX Knowledgebase ASCII release `ecotox_ascii_09_15_2026`, obtained from https://gaftp.epa.gov/ecotox/. The release contains 725,636 test records and 1,250,612 result records. We used it directly rather than through a curated derivative, so that the number of records per substance — the quantity at issue in this paper — reflects the primary compilation.

Selection proceeded in the order shown in Table S1. Tests were restricted to aquatic organisms in freshwater media; results were restricted to acute LC50, EC50 and IC50 and chronic NOEC, LOEC, LOEL, NOEL and MATC endpoints reported in mass-concentration units; and values carrying an inequality qualifier were separated out. Records in molar units (`uM`, `nM`, `mM`, `nmol/L`, `umol/L`) and in dose or body-weight units (`ppm diet`, `mg/kg bdwt`) were not converted and were excluded; molar units alone account for 3,967 of the rejected PFAS records.

**Table S1:** Record selection for the PFAS set.

| Step | Count |
|---|---|
| Tests on PFASSTRUCTV6 CAS numbers | 11,719 |
| after restricting to aquatic organisms in freshwater | 3,816 |
| Results joined to those tests | 29,375 |
| with an acute or chronic endpoint of interest | 24,379 |
| with convertible mass-concentration units and a numeric value | 16,357 |
| carrying an inequality qualifier (excluded) | 366 |
| **Usable uncensored records** | **15,991** |
| Unique substances represented | 103 |

The endpoint composition of the PFAS set is strongly asymmetric (Table S2) and this matters for the argument of the paper. Chronic sublethal endpoints outnumber acute endpoints four to one among distinct records (fifteen to one before duplicate rows are collapsed), and they also lie lower on the concentration scale. For 67 of the 87 parent structures (77%) the protective minimum is drawn from the chronic sublethal pool, and the size of that pool is what varies most between well- and poorly-studied substances.

The same filters applied without the PFAS restriction, and limited to acute endpoints, gave 106,095 freshwater acute records for 4,757 chemicals, which is the set used for the cross-chemical replication in Section S2 and as the training pool in Section S5.

**Table S2:** Endpoint composition of the PFAS records, before and after exact duplicate rows are collapsed. The acute pool (LC50, EC50, IC50) is 953 distinct records against 3,791 chronic, so a minimum taken over all strata is drawn from the chronic pool in the large majority of cases.

| Endpoint | Raw rows | Distinct records |
|---|---|---|
| NOEC | 8,584 | 1,918 |
| LOEC | 6,360 | 1,811 |
| EC50 | 523 | 514 |
| LC50 | 437 | 421 |
| NOEL | 43 | 43 |
| IC50 | 18 | 18 |
| MATC | 17 | 10 |
| LOEL | 9 | 9 |
| total | 15,991 | 4,744 |

## S2 Endpoint constructions and the effort diagnostic

All summaries are formed after structure standardisation and after exact duplicate rows are removed, so the unit of analysis is the parent structure and the unit of effort is the distinct record. Four summaries of the same records were compared, all expressed as $y = \log_{10}(T/\mathrm{mg\,L^{-1}})$: the minimum over all strata (the protective convention), the median over the same records, the median over acute records only, and a depth-equalised minimum described in Section S3. Testing effort was taken as $\log_{10}$ of the number of usable records per substance.

**Table S3:** Association between each endpoint and testing effort. No model is involved; this is a property of the response variable.

| Set | Endpoint | $n$ substances | Spearman $\rho$ vs $\log_{10}$ records |
|---|---|---|---|
| PFAS | minimum (all strata) | 87 | −0.749 |
| PFAS | minimum, effort counted as distinct tests | 87 | −0.703 |
| PFAS | median (all strata) | 87 | −0.364 |
| PFAS | acute median | 58 | −0.070 |
| PFAS | acute median vs $\log_{10}$ *acute* records | 58 | −0.132 |
| All chemicals | acute minimum | 4,757 | −0.462 |
| All chemicals | acute median | 4,757 | −0.160 |

The last two rows are the cross-chemical replication. Restricting to acute data does not by itself remove the confound: within acute records alone, the minimum still carries roughly three times the association of the median. The confound attaches to the act of taking a minimum, not to the inclusion of chronic data, although the two compound in the PFAS set because the chronic pool is both larger and lower.

Spread is the within-substance range in $\log_{10}$ units and rises with depth as expected for a sample range. The final column shows that the median is far less sensitive to depth than the minimum (a swing of 1.7 $\log_{10}$ units across the quartiles against 4.9 for the minimum), which is the basis for preferring it. Note also that the acute median's association is −0.070 against total records but −0.132 against the acute records that actually form it; on $n = 58$ neither differs significantly from zero, and we describe the acute median as effort-*stable*, not effort-free.

**Table S4:** Protective minimum by record-count quartile, PFAS parent structures. Quartiles are used rather than fixed cut-offs so that every bin is populated.

| Records per substance | $n$ | Median minimum | Median spread | Median of medians |
|---|---|---|---|---|
| 1–2 (Q1) | 26 | +1.16 | 0.00 | +1.25 |
| 3–8 (Q2) | 19 | −0.52 | 1.03 | +0.11 |
| 9–29 (Q3) | 20 | −2.00 | 3.14 | −0.15 |
| 30–1548 (Q4) | 22 | −3.76 | 6.04 | −0.42 |

## S3 Equal-depth subsampling

Every substance with at least $k$ records was subsampled to exactly $k$ records without replacement, the minimum recomputed, and the correlation with the original record count recalculated, over 400 draws per $k$.

The baseline matters. A substance is eligible for depth $k$ only if it has at least $k$ records, so the eligible set shrinks and shifts toward the well-studied substances as $k$ rises. Comparing the depth-equalised correlation with the whole-set value therefore mixes a change of sample with the effect of depth. We report both: the correlation over the eligible substances using *all* of their records (the baseline), and the correlation over the same substances at depth $k$. Their difference is the depth effect.

**Table S5:** Depth equalisation against the correct baseline. The depth effect is stable in $k$; the apparent anomaly at $k = 20$ in a whole-set comparison is an artefact of the shifting eligible set.

| $k$ | Eligible | Baseline $\rho$ | Depth-equalised $\rho$ | Depth effect |
|---|---|---|---|---|
| all records | 87 | −0.749 | — | — |
| 3 | 61 | −0.700 | $-0.338 \pm 0.071$ | 0.362 |
| 5 | 52 | −0.642 | $-0.329 \pm 0.078$ | 0.313 |
| 10 | 38 | −0.666 | $-0.370 \pm 0.083$ | 0.296 |
| 20 | 28 | −0.645 | $-0.343 \pm 0.100$ | 0.301 |

The depth effect is 0.30 to 0.36, i.e. 40–48% of the full-sample $\rho = -0.749$, and does not drift with $k$.

### Why the residual is not identified as reverse causation

Subsampling removes the effect of sampling *depth*. It cannot remove the effect of sampling *breadth*, because it draws from each substance's own record pool and therefore preserves that substance's dispersion and assay mix by construction. Breadth is strongly effort-dependent in these data:

**Table S6:** Within-substance dispersion rises with testing effort ($n = 61$ substances with at least three records). Spearman correlation against $\log_{10}$ record count.

| Quantity | $\rho$ | $p$ |
|---|---|---|
| within-substance standard deviation | +0.585 | $7 \times 10^{-7}$ |
| within-substance range | +0.818 | $8 \times 10^{-16}$ |
| distinct endpoint types reported | +0.743 | $7 \times 10^{-12}$ |

A draw of $k$ records from a more dispersed distribution still has a lower minimum, and subsampling leaves that channel intact. The residual $\rho \approx -0.34$ is therefore a residual, not an identified causal quantity: it is consistent with reverse causation, with effort-dependent assay breadth, and with the compositional shift toward chronic endpoints, and this design cannot separate them. Earlier versions of this analysis described the residual as reverse causation; that description was not warranted and has been withdrawn.

## S4 Fixed-model comparison across endpoint constructions

The broad non-PFAS model of the published analysis was held fixed and only the external target was changed, so that differences cannot be attributed to training, tuning or representation. All targets are computed on the parent-collapsed, de-duplicated record set.

**Table S7:** External performance of one fixed model against four constructions of the same underlying records. $R^2_{\mathrm{bc}}$ removes a single fitted intercept; $R^2_{\bar{y}}$ is the trivial predictor that returns the broad training mean; $s$ is the ratio of predicted to observed standard deviation. Confidence intervals are substance-level bootstrap (3,000 resamples).

| Endpoint | $n$ | $R^2$ | $R^2_{\mathrm{bc}}$ | $R^2_{\bar{y}}$ | $\rho$ | 95% CI on $\rho$ | $\rho$ vs effort |
|---|---|---|---|---|---|---|---|
| minimum (published) | 87 | $-0.714$ | $+0.011$ | $-0.358$ | $-0.020$ | $[-0.24, +0.18]$ | $-0.749$ |
| median | 87 | $-0.161$ | $-0.010$ | $-0.001$ | $+0.048$ | $[-0.18, +0.27]$ | $-0.364$ |
| depth-equalised minimum | 52 | $-3.348$ | $-0.036$ | $-1.823$ | $+0.171$ | $[-0.14, +0.44]$ | $-0.405$ |
| acute median | 58 | $+0.247$ | $+0.325$ | $-0.386$ | $+0.412$ | $[+0.18, +0.60]$ | $-0.070$ |

Three things follow and all three qualify the headline. First, the $R^2$ column is dominated by location: the minimum sits $+2.15$ $\log_{10}$ units from the training target's mean, and removing one intercept moves its $R^2$ from $-0.714$ to $+0.011$. The $R^2$ collapse therefore largely restates the order-statistic shift rather than providing independent evidence about predictive structure. Second, the model's dynamic range is narrow on every target ($s = 0.33$ to $0.65$), so $R^2$ is additionally penalised against dispersed targets. Third, the rank column survives both objections.

### Matched-sample comparison

The four constructions are not defined on identical substance sets: an acute median exists only for substances carrying acute records. The decisive comparison is therefore made on the 58 parent structures common to the minimum and the acute median.

**Table S8:** The same 58 parent structures, the same fixed model, two endpoints. The difference is a paired bootstrap over 4,000 resamples.

| Endpoint | $n$ | $R^2$ | $\rho$ (95% CI) |
|---|---|---|---|
| minimum | 58 | $-0.650$ | $+0.086$ $[-0.18, +0.35]$ |
| acute median | 58 | $+0.247$ | $+0.412$ $[+0.18, +0.60]$ |
| difference | | | $+0.321$ $[+0.08, +0.57]$, $p = 0.005$ |

The depth-equalised minimum is an informative intermediate: its rank correlation improves over the raw minimum, consistent with removing part of the confound, while its $R^2$ is very poor.

We previously attributed that to compressed variance, which is only part of the story: the depth-equalised target's variance is 2.03 against the acute median's 2.45, so it is indeed slightly compressed, but the dominant term is location — averaging subsampled minima shifts the target +2.59 $\log_{10}$ units from the model's training mean, and 76% of its mean squared error is that bias. We report it as a diagnostic, not as a candidate endpoint.

## S5 Consistent rebuild

Training and external sets were rebuilt on the acute median. Model settings were those of the published analysis and were not re-tuned: 2,048-bit ECFP4 fingerprints (Morgan radius 2), random forest with 300 trees, `max_features` $= \sqrt{p}$, seed 20260620. Broad cross-validation used five repeats of shuffled five-fold splitting with 160 trees per fold. Conformal validation used 40 repeats of the normalised split-conformal procedure given in the main text, with marginal calibration on a random 80/20 split of the broad set and PFAS-conditional calibration on disjoint halves of the PFAS set.

**Table S9:** The published study versus the present rebuild. The learner, representation and conformal procedure are identical, but the left column is the earlier study in full — a different source database (a curated EnviroTox derivative rather than ECOTOX), a different substance list and a different training set. It is *not* an endpoint-only contrast; Tables S7 and S8 are. Readers should not attribute the differences below to the endpoint alone.

| Quantity | Protective minimum | Acute median |
|---|---|---|
| Broad training structures | 3,788 | 3,718 |
| External PFAS structures | 138[1] | 58 |
| Broad $5 \times 5$ CV $R^2$ | $0.387 \pm 0.024$ | $0.443 \pm 0.029$ |
| External $R^2$ | $-0.580$ | $+0.366$ |
| External Spearman $\rho$ | $-0.071$ | $+0.460$ |
| 95% CI | — | $[+0.23, +0.65]$ |
| Marginal coverage at nominal 0.90 | 0.659 | 0.851 |
| Marginal width | 4.84 | 3.70 |
| PFAS-conditional coverage | 0.922 | 0.897 |
| PFAS-conditional width | 10.53 | 4.48 |
| Inventory structures in domain | 17,128 | 16,070 |

Within the present extraction the external set falls from 87 parent structures on the protective minimum to 58 on the acute median, because 29 substances carry only chronic records. That reduction is deliberate, and the smaller external set is the principal limitation of the rebuilt analysis.

Per-repeat conformal results are deposited rather than summarised only by their means. Over the 40 calibration splits, marginal coverage was 0.851 (2.5th–97.5th percentile $[0.793, 0.897]$) with width 3.70 $[3.40, 4.00]$, and PFAS-conditional coverage 0.897 $[0.724, 1.000]$ with width 4.48 $[3.34, 5.35]$. Because each repeat re-uses the same 58 external structures, the effective sample for coverage is 58 (29 for the conditional procedure), not $40 \times 58$; the corresponding exact binomial intervals are $[0.726, 0.927]$ and $[0.726, 0.978]$. Neither coverage differs significantly from nominal, and neither could the design have distinguished from 0.81.

Marginal calibration splits the broad non-PFAS set and tests on PFAS. Calibration and test are

[1] From the superseded EnviroTox-derived extraction (138 measured substances, of which 119 remain after desalting with neutralisation); the external statistics in that column were computed on the 138. It is not a subset of the 87 parent structures of the present ECOTOX extraction.

therefore not exchangeable and the split-conformal guarantee does not apply to it; we report it for comparability with the published analysis and deploy only the PFAS-conditional procedure. The deployment intervals on the untested inventory inherit a further caveat: they are calibrated on the 58 *measured* PFAS, which are by this paper's own argument the well-studied end of the distribution, so exchangeability with the untested inventory is not established and the nominal 90% should be read as a construction rather than a guarantee.

## S6 Clearance across hazard thresholds

Because the two endpoints differ in protectiveness, a single threshold does not provide a comparable decision rule. On the 58 parent structures carrying both summaries the acute median lies above the protective minimum by +2.66 $\log_{10}$ units in the mean and +2.32 in the median, and 93.1% of measured PFAS exceed 0.1 mg/L on the acute median against 39.7% on the minimum.

A "stringency-matched" threshold is a construction and is not unique. Shifting 0.1 mg/L by the mean offset gives 46 mg/L, by the median offset 21 mg/L, and matching instead on the fraction of measured PFAS above the cut-off (39.7% lie above 0.1 mg/L on the minimum, so the matching value is the 60.3rd percentile of the acute median) gives 41 mg/L. Clearance is zero at all three, and at every threshold at or above 10 mg/L, so no conclusion in this paper depends on the choice. We also emphasise that matching the fraction of substances flagged matches the *selectivity* of a screen, not the protection it delivers, and that none of these values is a threshold any regulatory framework applies. Clearance was evaluated across a continuum for that reason.

**Table S10:** Clearance of in-domain inventory structures ($n$ = 16,070) under the rebuilt model, by hazard threshold. The required half-width is the median over in-domain structures of the distance from the point prediction to the threshold; the achieved median half-width is 2.44.

| Threshold | Cleared | % | Median required half-width |
|---|---|---|---|
| 0.01 mg/L | 14,991 | 93.3 | +3.22 |
| 0.1 mg/L (used with the protective endpoint) | 5,441 | 33.9 | +2.22 |
| 1 mg/L (GHS acute aquatic Category 1) | 306 | 1.9 | +1.22 |
| 10 mg/L | 0 | 0.0 | +0.22 |
| 21 mg/L (median-offset match) | 0 | 0.0 | −0.11 |
| 46 mg/L (mean-offset match) | 0 | 0.0 | −0.45 |
| 41 mg/L (exceedance-fraction match) | 0 | 0.0 | −0.39 |
| 100 mg/L | 0 | 0.0 | −0.78 |

Two regimes are visible. Above 10 mg/L the required half-width approaches and then falls below zero, meaning the median in-domain structure is predicted more hazardous than the threshold and no narrowing of the interval would clear it. Below that, the barrier is interval width alone: at the GHS Category 1 boundary the median structure would clear with a 90% interval of 2.43 $\log_{10}$ units against the 4.87 achieved.

## S7 Structure standardisation and resolution

ECOTOX identifies substances by CAS number and carries no structures. CAS numbers were converted to the dashed form and resolved to SMILES through the PubChem PUG REST interface at the published rate limit of five requests per second; 4,311 of 4,757 chemicals resolved (90.6%),

and the remainder were dropped. Resolution failures are not random with respect to chemical class — mixtures, poorly defined substances and some organometallics resolve less often — and we were not able to characterise the resulting bias.

Two standardisation choices differ from common practice and are stated explicitly because they change the data.

*Desalting keeps the fluorine-bearing fragment, not the largest one.* The usual rule retains the largest fragment by heavy-atom count. For tetrabutylammonium perfluoro-1-butanesulfonate the tetrabutylammonium cation (17 heavy atoms) exceeds the perfluorobutanesulfonate anion (13 heavy atoms), so the largest-fragment rule discards the analyte and models the counterion. We therefore retain the fluorinated fragment where one exists, falling back to the largest fragment otherwise.

*Fragments are charge-neutralised before canonicalisation.* Without this step a free acid and its salts produce different canonical SMILES and are counted as distinct structures. In the present ECOTOX extraction, 103 CAS registrations collapse to 87 parent structures under desalting with neutralisation. The difference is not cosmetic: ten registrations resolve to perfluorooctanesulfonate or its close derivatives, and their separately assigned protective values span 9.1 $\log_{10}$ units, from $-6.00$ for the potassium salt to $+3.13$; without collapsing they enter the analysis ten times, at ten different effective testing depths, and can fall on both sides of a calibration split.

*Standardisation precedes summarisation.* The order matters and an earlier version of this work had it wrong: the per-substance table was formed from CAS registrations and the structure rule applied only downstream, so the primary effort analysis ran on 103 uncollapsed registrations rather than 87 parents. The table is now built by pooling the records of every registration mapping to one parent and computing the endpoint once on the pooled set. Correcting it strengthens the central result — the effort correlation moves from $-0.734$ on the uncollapsed registrations to $-0.749$ on parent structures — and we record the correction rather than only the corrected number.

*Exact duplicate rows are removed before counting effort.* The PFAS extraction returns 15,991 rows, of which only 4,744 are distinct in test identifier, endpoint type, reported concentration and unit; these arise from multiple result records attached to one test. Because testing effort is measured by counting records, duplicated rows inflate it, and unevenly. Deduplication leaves the finding intact: $\rho = -0.749$ on distinct records and $-0.703$ when effort is counted as distinct tests (2,182) instead.

## S8 OECD subclass assignment and statistical power

Substances were assigned to structure categories following the OECD (2021) definition and the 2018 OECD/UNEP structure series. The definitional gate requires at least one fully fluorinated, saturated methyl or methylene carbon carrying no hydrogen, chlorine, bromine or iodine. Carbons that are fully halogenated but not fully fluorinated ($CHF_2$, $CClF_2$), methine carbons (>CF–) alone, and unsaturated carbons (=$CF_2$) do not satisfy it.

Applying the gate to the 87 parent structures of the present extraction excluded four: 1,1,2-trichloro-1,2,2-trifluoroethane (non-fluorine halogens on the candidate carbon), 2,3,4,5,6-pentafluoroaniline and pentafluorobenzaldehyde (fluorine attached only to an aromatic ring), and difluoroacetic acid, whose $CHF_2$ carbon bears a hydrogen. Eighty-three substances pass.

Categories were assigned by parsing the molecule as $C_nF_{2n+1}$–X–R with head-group identity taking precedence over incidental fluorination. Assignment uses connectivity from fluorinated carbons rather than requiring a literal $CF_2$ bonded to the head group, so that perfluoroalkyl ether acids (for

example HFPO-DA, whose $\alpha$-carbon is a methine) and perfluorinated cycloalkane sulfonic acids are classified by head group rather than falling through to a residual category.

## Corrections to the classifier

Four defects in the cascade used in an earlier version of this work have been corrected, and we list them because each produced visibly wrong assignments. (i) The carbonyl search walked one bond from a fluorinated carbon, so an $\alpha,\beta$-unsaturated perfluoroalkenoic acid — whose acid carbon sits two bonds away across a =CH– that bears hydrogen — was invisible and the molecule fell to the residual category; the walk now crosses one $sp^2$ carbon. (ii) The carbonyl test did not distinguish an acid from a ketone, ester or acyl halide, so perfluoro-2-methyl-3-pentanone and ethyl perfluoropentanoate were assigned to the carboxylic-acid category; an acid now requires a second oxygen bearing hydrogen or negative charge. (iii) Category 700 was assigned whenever any aromatic ring was present and nothing else matched, which placed tetraconazole — fluorinated on an ether side chain, with a dichlorophenyl ring bearing no fluorine — in a "side-chain fluorinated aromatic" class; the fluorinated carbon must now lie within two bonds of an aromatic atom. (iv) The fluorotelomer SMARTS was satisfied by any $C_nF_{2n+1}CH_2CH_2$– unit with $n \geq 2$, which classified fulvestrant, a steroidal antioestrogen carrying a $C_2F_5$ terminus, as a telomerisation product; the floor is now $n \geq 4$, matching the 4:2, 6:2, 8:2 and 10:2 series of the 2018 OECD/UNEP categories. The cost of (iv) is that 3:1 fluorotelomer alcohol is assigned by head group instead; we accept that in exchange for excluding pharmaceuticals.

## Class-resolved transfer

Classification is performed on the acute median, the endpoint this paper endorses; a class-resolved analysis of the protective minimum would inherit the effort confound one class at a time. Fifty-six of the 83 gate-passing substances carry acute records. All nine categories are listed; those with fewer than five members are not estimated, because a bootstrap interval on three points is degenerate.

**Table S11:** External rank correlation by OECD structure class on the acute median. Intervals are substance-level bootstrap (4,000 resamples). The detectable $|\rho|$ is at 80% power and $\alpha = 0.05$.

| Class | $n$ | Spearman $\rho$ | 95% CI | Detectable $\lvert\rho\rvert$ |
|---|---|---|---|---|
| All OECD-2021 PFAS | 56 | +0.412 | [+0.18, +0.61] | 0.37 |
| 400 fluorotelomer | 9 | −0.439 | [−0.91, +0.53] | 0.82 |
| 200a perfluoroalkane sulfonic | 7 | −0.134 | [−0.91, +0.91] | 0.89 |
| 200b perfluoroalkane sulfonamide | 6 | −0.116 | [−1.00, +0.80] | 0.92 |
| 700 side-chain fluorinated aromatic | 8 | +0.214 | [−0.60, +0.84] | 0.85 |
| 100 perfluoroalkyl carboxylic | 13 | +0.569 | [−0.05, +0.90] | 0.71 |
| 600 other perfluoroalkyl | 7 | +0.679 | [−0.19, +1.00] | 0.89 |
| *Listed but not estimated ($n < 5$):* | | | | |
| 500 perfluoroalkyl ether acid | 3 | — | — | — |
| 500 perfluoroalkyl ether other | 3 | — | — | — |
| 300 perfluoroalkyl phosphorus | 0 | — | — | — |

The class sizes sum to 56, which is the number of gate-passing substances carrying acute records; no category is omitted.

Every estimable class interval covers zero. The pooled estimate does not. At these sizes 80% power detects only $|\rho| \geq 0.71$ to 0.92, so the point estimates carry essentially no information about differential transfer.

### Correction to a previously reported positive result

An earlier version of this analysis reported perfluoroalkyl ether acids at $\rho = +0.695$ (95% CI $[+0.14, +0.93]$, $n = 13$) as the one class excluding zero. That analysis was computed on the superseded EnviroTox-derived external set and on the protective endpoint, and does not survive recomputation: in the present extraction the category carries three substances with acute records and is not estimable. We record the correction here rather than silently dropping it.

We also tested whether class-level transfer merely tracks class-level testing effort, which would make the class analysis a third measurement of the confound rather than a statement about chemistry. It does not: across the six estimable classes the correlation between class rank correlation and class mean $\log_{10}$ record count is $\rho = -0.49$ ($p = 0.33$). The sign is the opposite of what the confound hypothesis predicts and the estimate is far from significant on six points, so we draw no positive conclusion from it either.

### What would be required

Resolving differential transfer at $\rho = 0.3$ requires $n \approx 85$ per class at 80% power, or 129 with Bonferroni correction for six comparisons. Across six categories that is roughly 510 to 780 measured PFAS, against the 87 parent structures the public record currently supports.

### A note on multiplicity

No multiplicity correction is applied to the intervals reported anywhere in this work; they are nominal 95% intervals. This is immaterial for the effort association ($p = 7 \times 10^{-17}$) and for the pooled transfer estimate, and it is stated here because the class table above reports six intervals and the reader should not treat any of them as a corrected test. Under Bonferroni correction for six comparisons no class interval would exclude zero either, so the conclusion is unchanged.

### Conformal procedure actually used

Equation (1) of the main text defines a normalised nonconformity score, and all conformal results reported here and in the deposited code use it. No leave-one-subclass-out conformal analysis is reported in this paper. An earlier exploratory version computed one using unnormalised absolute residuals, which is inconsistent with Equation (1); that analysis has been withdrawn and no number in this manuscript depends on it. We state this because the discrepancy existed in a prior draft.

## S9 Representation and learner, held against the endpoint

The endpoint is not the only choice that affects transfer, and we report the alternatives we examined rather than asserting that they do not matter. Both tables below hold the training data and the external set fixed and vary one thing.

**Table S12: Computed on the superseded EnviroTox-derived extraction** ($n_{\text{train}}$ = 3,788/3,438/1,714; external $n$ = 138/97/108), not on the 87-parent ECOTOX extraction used elsewhere in this work. We have not repeated it, and report it because it contradicts a claim we previously made. Learner and endpoint are fixed within each block; only the representation varies.

| Endpoint | Representation | $n$ | $R^2$ | Spearman $\rho$ |
|---|---|---|---|---|
| most sensitive | ECFP4 binary (as published) | 138 | −0.684 | −0.026 |
| most sensitive | Morgan count | 138 | −0.618 | −0.020 |
| most sensitive | RDKit 165 descriptors | 138 | −0.064 | +0.419 |
| acute | ECFP4 binary | 97 | −0.057 | +0.296 |
| acute | Morgan count | 97 | −0.023 | +0.317 |
| acute | RDKit 165 descriptors | 97 | +0.233 | +0.478 |
| chronic | ECFP4 binary | 108 | −0.677 | +0.017 |

This table qualifies the paper's argument and we state the qualification plainly, with the caveat that it is computed on the earlier extraction. On the protective minimum, replacing the binary fingerprint with 165 physicochemical descriptors moves rank transfer from $\rho = -0.026$ to $+0.419$ — a recovery of similar magnitude to the one obtained by changing the endpoint. Representation and endpoint are therefore both material, and an earlier version of this work asserted that the failure "is not repaired by a better representation". That assertion was contradicted by our own deposited results and has been withdrawn.

Two things keep the endpoint argument standing. The comparison above is made on each endpoint's full available set and on the earlier data, whereas the endpoint comparison in Table S8 is matched on 58 common structures from the present extraction; and a descriptor representation does not make the protective minimum an effort-stable quantity — it remains correlated with record count at $\rho = -0.749$ whatever the model is given as input. The claim we retain is that endpoint construction is the lever a screening programme controls when it defines its response variable, not that it is the only lever.

**Table S13:** Learner, with the representation and the protective endpoint fixed, on the superseded external set. Reported for completeness; no conclusion in this paper rests on it.

| Learner | External $R^2$ | Spearman $\rho$ |
|---|---|---|
| Random forest (as published) | −0.580 | −0.071 |
| Gradient boosting | −0.596 | −0.041 |
| Ridge regression | −0.280 | +0.086 |

A linear model outperforms both ensembles on this target. We do not read that as evidence about chemistry: on a target that is partly a function of testing depth, a lower-variance predictor is penalised less, which is what the bias-corrected columns of Table S7 also show.

## S10 Applicability domain: dependence on the distance metric

The applicability domain is the mean Euclidean distance to the five nearest training structures, with the threshold at the 95th percentile of the training distribution. Euclidean distance on a binary fingerprint counts shared *absences* among the roughly 2,025 unset bits as evidence of similarity;

Tanimoto similarity does not, and is the conventional choice in this field.

**Table S14:** Fraction of the untested PFAS inventory admitted to the applicability domain, by distance metric and percentile threshold. Computed on the 18,851-structure inventory of the earlier run; the rebuilt model places 16,070 of 17,619 structures (91.2%) in domain under the same Euclidean rule, so the comparison between metrics, not the absolute percentage, is what this table establishes.

| Threshold | Euclidean (as used) | Tanimoto |
|---|---|---|
| 90th percentile | 79.8% | 30.9% |
| 95th percentile | 90.9% | 51.6% |
| 99th percentile | 97.1% | 90.0% |

The median maximum Tanimoto similarity of an untested PFAS to any of the broad training structures is 0.27, and 86% fall below 0.40. By any conventional criterion the untested inventory lies largely outside the domain of a model trained on non-PFAS chemicals, and the 90.9% figure should be read as a property of the Euclidean rule rather than as a statement that these structures are well covered.

This does not affect the clearance conclusion. At any threshold at or above 10 mg/L the point prediction of the median in-domain structure is already on the wrong side, so no restriction of the domain changes the zero. It does affect the in-domain count, the median interval width and the 33.9% figure at 0.1 mg/L, which are correspondingly optimistic, and which we therefore do not present as deployment-ready quantities.

## S11 Files supporting the numerical audit

Release `v3.1` of https://github.com/bisu9082/pfas-clearance-validation. Scripts are listed in execution order; each entry names the tables and figures it produces.

- `extract.py` — record extraction from the ECOTOX ASCII release.
- `resolve_smiles.py` — CAS to SMILES resolution via PubChem, resume-safe.
- `recurate.py` — structure standardisation applied *before* summarisation, duplicate-row removal, and the per-substance table `ecotox_pfas_measured_recurated.csv` (Tables S3–S4).
- `subsample.py` — equal-depth subsampling against the eligible-subset baseline, and the dispersion/effort diagnostic (Tables S5–S6).
- `endpoint_fix2.py` — fixed-model comparison across endpoint constructions, bias-corrected $R^2$, and the matched-58 paired bootstrap (Tables S7–S8).
- `build_broad.py` — acute-median candidate set across all 4,757 chemicals.
- `final_pipeline.py` — consistent rebuild: cross-validation, external validation, conformal intervals, inventory clearance (Tables S9–S10).
- `oecd_classify.py`, `run_oecd.py` — OECD (2021) definitional gate and category assignment, with the four classifier corrections of §S8.
- `forest2.py` — class-resolved transfer on the acute median (Table S11).

- `make_fig01.py`, `make_fig02.py`, `make_fig03.py` — Figures 1–3. Each ends in a measured-extent layout assertion; `make_fig03.py` additionally asserts that every tick label sits on the row of the class it names.

Result files, in `canonical_results/`:

- `ecotox_pfas_records.csv.gz` — extracted freshwater records.
- `ecotox_pfas_measured_recurated.csv` — the 87 parent structures.
- `recuration_metrics.json`, `subsample.json` — Tables S3–S6.
- `endpoint_fix_recurated.json` — Tables S7–S8.
- `oecd_103set.csv`, `forest_final.csv` — Table S11.
- `rebuild_v2_conformal_repeats.csv` — per-repeat conformal results.
- `final_inventory_predictions.csv`, `clearance.json` — Table S10.

Superseded files from the earlier study are retained under `archive_v2/` with a notice and are the source of Tables S12–S14; no other table in this document draws on them.